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Partnering Science and Art

Arts and STEM partnerships in the primary classroom

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Draw what you see



Close-up of the satellite image from the science and art unit, University of Wyoming

- 90 seconds: draw the river and the ridges around it
- Draw only what you can see
- Tell your group of three one thing you noticed only because you drew it

Observe · Represent · Test · Revise

Science and art share a working method: look closely, represent what you see, test it, and revise.

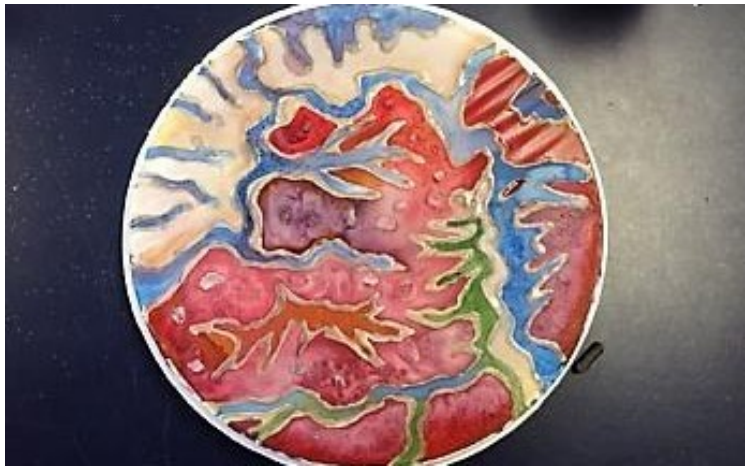
Practices that pair

Science practice	Studio habit of mind
Observe and record	Observe
Question and predict	Envision
Use equipment with skill	Develop craft
Investigate, then try again	Engage and persist · stretch and explore
Communicate findings	Express
Evaluate evidence	Reflect

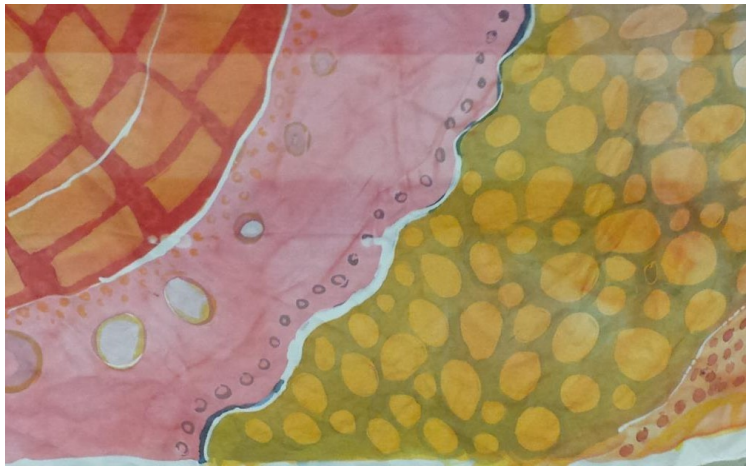
After one shared unit, our pre-service teachers named the same overlaps: problem solving, observation, and visual representation

Studio habits of mind: Hetland, Winner, Veenema, and Sheridan (2013) · overlaps: Belardo, Burrows, and Dambekalns (2017)

Science and art methods courses, joined for five sessions



The satellite image, painted on a silk sampler



Detail of a group batik of a microscopic image

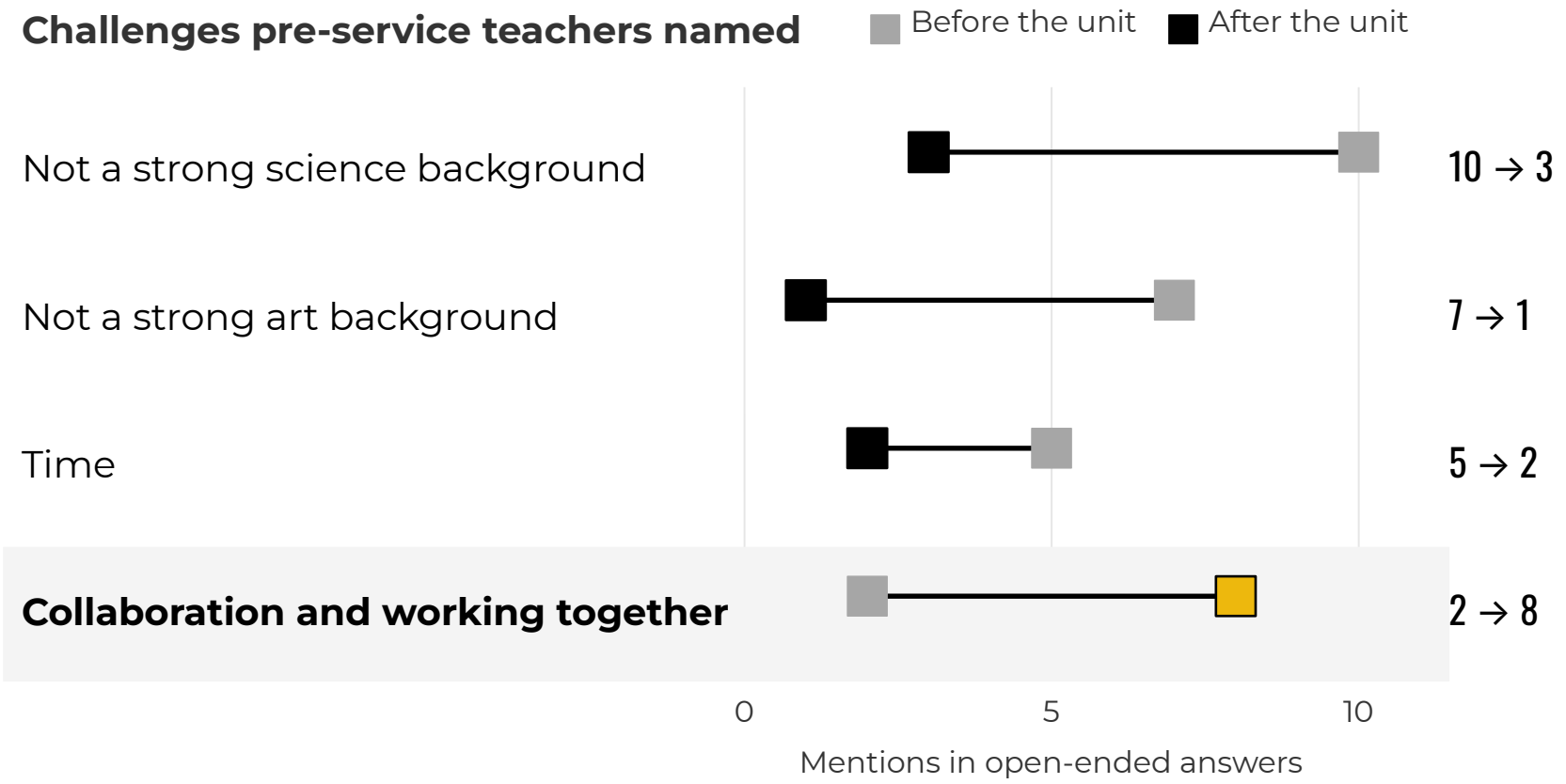


Presenting the finished batik to the class

Micro-teach	**Sampler**	**Agree**	**Paint**	**Present**
Each one, teach one: a science or art idea	A satellite image on a small silk round	Choose one microscopic image together	A group silk batik, about 90 × 120 cm	Science and art ideas, then critique

University of Wyoming · 3 cohorts, 2012–2014 · 52 secondary pre-service teachers · mixed teams of about four, two from each subject · about 7 hours

The worry moved from my subject to our partnership



7 of 8

confidence items rose significantly in every cohort

6.0–6.8 → 8.2–8.8

cohort mean confidence, out of 10

Three cohorts combined, 52 pre-service teachers · Belardo, Burrows, and Dambekalns (2017), Problems of Education in the 21st Century

Partnerships grow through three levels

Dimension	Pre-partnership	partnership (little p)	Partnership (big P)
Focus	Getting to know each other	Working toward goals both value	Developing and carrying out the work together
Activities	Limited: getting acquainted	Opportunistic: a convenient match	Integral: grow from shared skills and interests
Benefit	New contacts and skills	More capacity than either has alone	The team becomes more than it would be alone
Trust and respect	Building trust, earning respect	Present among some members	Mutual, throughout the team
Influence	Separate	Shared or divided by expertise	Integrated; expertise acknowledged

Teams usually move left to right; a team can sit at different levels on different dimensions

Kilty and Burrows (2022), Education Sciences 12, 58, adapted from Mullinix (2001) · five of the model's dimensions shown

Better partnerships, better products





Launch and recovery, LIFT program, University of Wyoming

- Teams of three: one engineering, one science, and one education major
- Each team built a high-altitude balloon payload and taught it to school pupils
- Every team reached partnership; benefit and products reached Partnership
- The stronger the partnership, the better the product

Your partnerships pre, p, or P? Which dimensions are you using, and on purpose?

Kilty and Burrows (2022): 9 undergraduates in 3 teams, interviews and field notes · LIFT: Learning to Integrate Fundamentals through Teaching

How deep does the subject partnership go?

Decoration

Art dresses up the science. Only the science is taught or assessed.

Integration

One task teaches and assesses both subjects.

Transdisciplinary

A shared problem that neither subject solves alone.

Deeper partnership between the subjects

After Liao (2016) and Perignat and Katz-Buonincontro (2019)

The partnership test

Remove the art. If the science learning stays the same, the art was decoration.

Four partnered lessons to borrow

	Sound and music	Light and circuits	Forces and structures	Hidden worlds
Science	Vibration and pitch	Complete and broken circuits	Balance, stability, and load	Structure and classification
Art	Compose and perform a four-beat pattern	Composition: light as the focal point	Form, line, and balance in sculpture	Pattern, texture, and abstraction
Pupils make	Straw pan pipes or a micro:bit instrument	A light-up card: copper tape, one LED, a coin cell	A Strawbees or PIKS sculpture that carries a load	A digital microscope gallery of leaves, fabric, and salt
Assess both	Explains why one pipe sounds lower; performs the pattern	Traces the circuit; light leads the eye where intended	Holds the load; a two-sentence artist statement	Sorts images by structure; curates and titles a gallery

Materials from the CRAFT Maker Space at MIC · coin cells are a swallowing hazard: count them out and back

Your turn: plan one partnered lesson

- Groups of three · 12 minutes
- Pick a topic card
- Write a science outcome and an arts outcome
- Name a partner and the level you would aim for
- Apply the partnership test before you finish

Partnered lesson planner

Class level _____ Topic _____

Science outcome	
Arts outcome	
Shared practice	
What pupils make	
How you assess the science	
How you assess the art	
Partner, and the level you aim for (pre, p, or P)	
Partnership test: what is lost if the art is removed?	

Topic cards

Seasons and weather	Habitats	Light and shadow	Sound
Materials	Forces	Plant life cycles	Space

- Arts to pair: visual arts, music, drama, and dance

Share-out · 6 minutes

Share: what would be lost if the art were removed, and who would you plan it with?

Take it to placement

- Add an arts outcome you will assess to one science lesson you already teach
- Plan it with a partner, and agree in writing who does what
- Run the partnership test when you plan
- Borrow kits from the CRAFT Maker Space

Further reading

- Belardo, C., Burrows, A. C., & Dambekalns, L. (2017). Partnering science and art: Pre-service teachers' experiences for use in pre-collegiate classrooms. Problems of Education in the 21st Century, 75(3), 215–234.
- Kilty, T. J., & Burrows, A. C. (2022). Integrated STEM and partnerships: What to do for more effective teams in informal settings. Education Sciences, 12(1), 58.
- Hetland, L., Winner, E., Veenema, S., & Sheridan, K. M. (2013). Studio thinking 2 (2nd ed.). Teachers College Press.
- Liao, C. (2016). From interdisciplinary to transdisciplinary: An arts-integrated approach to STEAM education. Art Education, 69(6), 44–49.
- Perignat, E., & Katz-Buonincontro, J. (2019). STEAM in practice and research: An integrative literature review. Thinking Skills and Creativity, 31, 31–43.

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